

Lab 3: Selecting a model for Balance on a QNB credit card

70-467 - Machine Learning for Business Analytics

Andrés Castaño Zuluaga

2026-09-20

Background. We have been using the Credit data in class. Balance is the average amount on the card, in dollars. For this lab, let's assume these 400 accounts are credit-card balances on QNB (Qatar National Bank) cards.

QNB has already issued the cards. You have income, credit limit, credit rating, number of cards, age, years of education, whether the person owns a home, is married, and region, among other variables in the dataset. Mean Balance is about \$520. Ninety accounts sit at zero. Region here is East, South, or West. The codebook on the next page gives type and unit for every column.

The problem. We want a good predictive model for Balance on these QNB cards. There are many variables, so there are many possible models. If we put every column into one least-squares fit, or we pick columns because they look important, the fit on these 400 accounts can look better and better as we add variables. That can be overfitting. It does not mean the model will predict well on a new customer.

This is the same question we asked in class: how can we select a good predictive model for Balance among many, and how can we avoid overfitting?

What we will do in the lab. We will use the selection methods from class to work that question.

First we will search: best subset, forward stepwise, and backward stepwise. That gives you the best model of each size.

Then we will pick a size in the two ways we discussed. C_p , AIC, BIC, and adjusted R^2 adjust the training error. A validation sample or cross-validation scores customers you did not use to pick columns. Use both. They will not always pick the same size.

Do not pick columns from the p -values of the model that uses every column. We saw in class why that fails when columns move together.

Read the codebook before class. Bring a laptop. After class you will have additional questions connected to the selection slides.

Codebook. 400 rows, 11 columns. One row = one credit-card account. Source: ISLR2 `Credit` (James, Witten, Hastie, and Tibshirani). For this lab, credit-card balances on QNB cards, in dollars.

Variable	Type	Unit	Description
Balance	numeric	dollars	Average credit-card balance. The outcome.
Income	numeric	\$000s	Income in thousands of dollars.
Limit	numeric	dollars	Credit limit on the card.
Rating	numeric	points	Credit rating.
Cards	numeric	count	Number of credit cards.
Age	numeric	years	Age of the customer.
Education	numeric	years	Years of education.
Own	factor	Yes / No	Home owner.
Student	factor	Yes / No	Student.
Married	factor	Yes / No	Married.
Region	factor	none	East, South, or West.